

46-921, Fall 2020: Homework #4

Due 9:30 AM EDT, Thursday, October 1

Note: When you are asked to approximate the distribution of an estimator, it is **not** sufficient to just say “It is approximately normal.”

1. Let $X_1, X_2, \dots, X_n$ be i.i.d. from the *Pareto distribution*:

$$f_X(x) = \begin{cases} \frac{\alpha \lambda^\alpha}{x^{\alpha+1}} & x > \lambda, \alpha > 0, \lambda > 0 \\ 0 & \text{otherwise} \end{cases}$$

For this exercise we will assume that λ is a known constant.

- (a) Find the MLE for α , call it $\hat{\alpha}$.
 - (b) Approximate the distribution of $\hat{\alpha}$.
 - (c) A sample of size 100 is taken from a population that we are willing to assume has the Pareto distribution with $\lambda = 2$. It holds that $\sum_i \log(x_i) = 107.39$. Make a statement regarding our best estimate of α and attach a standard error to the estimator.
2. Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. from the $\text{Gamma}(\theta, \beta)$ distribution, where θ is unknown and β is known. Even though there is not a closed form for the MLE for θ , one can do the following:
 - (a) Derive the Fisher Information $I(\theta)$.
 - (b) Approximate the distribution of the MLE for θ .
 - (c) Approximate the distribution of the MLE for $\log(\theta)$.
 3. Suppose that X is $\text{binomial}(n, p)$. The MLE for p is, not surprisingly, the sample proportion $\hat{p} = X/n$. (You do not need to prove this.)
 - (a) What is the MLE for $p/(1-p)$? (This is called the *odds*.)
 - (b) Approximate the distribution of the MLE of the odds.
 - (c) Use part (b) to construct a $100(1-\alpha)\%$ confidence interval for the odds.
 - (d) Write a simulation procedure that tests the validity of the confidence interval found in part (c). Is the confidence interval an adequate approximation when $n = 10$ and $p = 0.10$?